

Building Resilient Supply Chains for Natural Disasters

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Group 11: Introduction to SCM: 33:799:301:08
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Agenda

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Framework for Disaster-Resilient Supply Chains

Natural Disaster Risks and Their Effects on the Supply Chain

Natural Disaster Risks:

Natural disasters are composed of geophysical, meteorological, and hydrological events. These include earthquakes, volcanic eruptions, hurricanes, floods, etc.

Natural disasters pose severe threats to supply chains. As supply networks have grown more complex, the blast radius of localized disasters have expanded globally.



Effects on Supply Chains:

First order effects: The immediate damage from the epicenter of the event. These include loss of inventory, equipment damage, and facility condemnation.

Second order effects: The disruption spreads beyond the impact zone. These include transportation bottlenecks, utility blackouts, and telecommunications failures.

Third order effects: The delayed reactions to the event, driven by the bullwhip effect. These include demand surges, panic buying, and cost inflation.

Examples of Major Disruptions Caused by Natural Disasters

Major Natural Disaster

Disruption

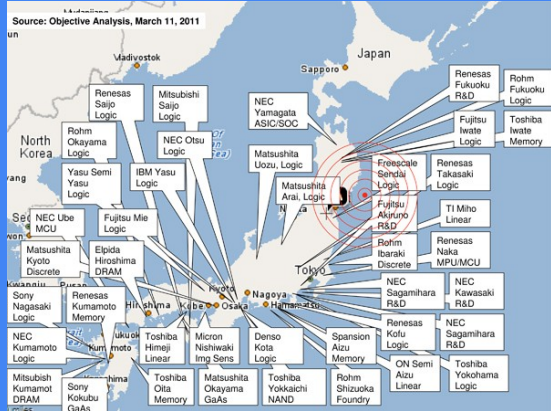


Hurricane Katrina (2005, USA)

- Hurricane Katrina shut down the Port of New Orleans, disrupting a major trade route
- It reduced U.S. oil supply and caused fuel shortages
- Transportation damage led to nationwide logistics delays

Japan Earthquake & Tsunami (2011)

- The disaster shut down key automotive and semiconductor factories.
- It caused shortages of critical components like chips
- Companies such as Toyota and Sony faced production delays
- The just-in-time system failed due to lack of inventory



Supply Chain Impact Analysis

Common Impacts:

- Natural disasters damage infrastructure such as ports, roads, and warehouses
- They disrupt suppliers and delay transportation flows

Business Consequences:

- These disruptions lead to inventory shortages and production stoppages
- They also increase operational costs for companies

Key Insight:

- Local disasters can quickly spread through global supply chains
- Overdependence on a single supplier increases vulnerability

Resilience Strategies

Multi-Sourcing

- Having multiple qualified suppliers for key materials in order to avoid dependence on a single supplier
- Spreading your suppliers across different regions in order to reduce your exposure to localized natural disasters
- Rapidly shifting your orders when one supplier is disrupted
- Ensuring that your suppliers use different production techniques so a disruption affecting one technique doesn't affect all of them

Geographic Risk Mapping

- Using historical disaster patterns and data to identify which suppliers, facilities, and transportation routes sit in high-risk geographic zones
- Assessing how parts of the supply chain are exposed to natural hazards in order to pinpoint the most vulnerable locations
- Redesigning the supply chain in order to shift critical operations from disaster-prone regions to lower-risk areas
- Positioning your inventory, capacity, and key logistics in safer locations ahead of seasonal disaster risks to maintain continuity

Buffer Inventory

- Maintaining additional safety stock of critical materials in advance for short-term disruptions
- Distributing your inventory across multiple warehouses so a single facility outage doesn't halt company operations
- Providing enough extra inventory to keep operations running while backup suppliers or logistics changes take effect
- Reducing reliance on emergency sourcing by ensuring essential inputs are available during shutdowns

Examples of Companies Effectively Managing Disaster Risks

Toyota

- 2011 Earthquake and tsunami
 - 77% (~\$1.4B) fall in profits
 - Shortage of Japanese manufactured parts → 30% production drop in North America
- Revised Toyota Production Plan
 - Standardized parts
 - Multi-sourcing
 - Required suppliers to hold inventory
- 2016 Earthquakes
 - Resumed production in 2-4 weeks
 - Lost limited to ~\$277 M



Apple

- Thai floods in 2011
 - Global hard-drive shortage → Risk to PC manufacturers, including Apple
- Apple Response
 - Just-In-Time model → supply chain visibility
 - Identified shortages & shifted production
- As a result
 - Minimal disruption/delays to Macbooks
 - Stable share price + Mac sale growth



Framework for Disaster- Resilient Supply Chains

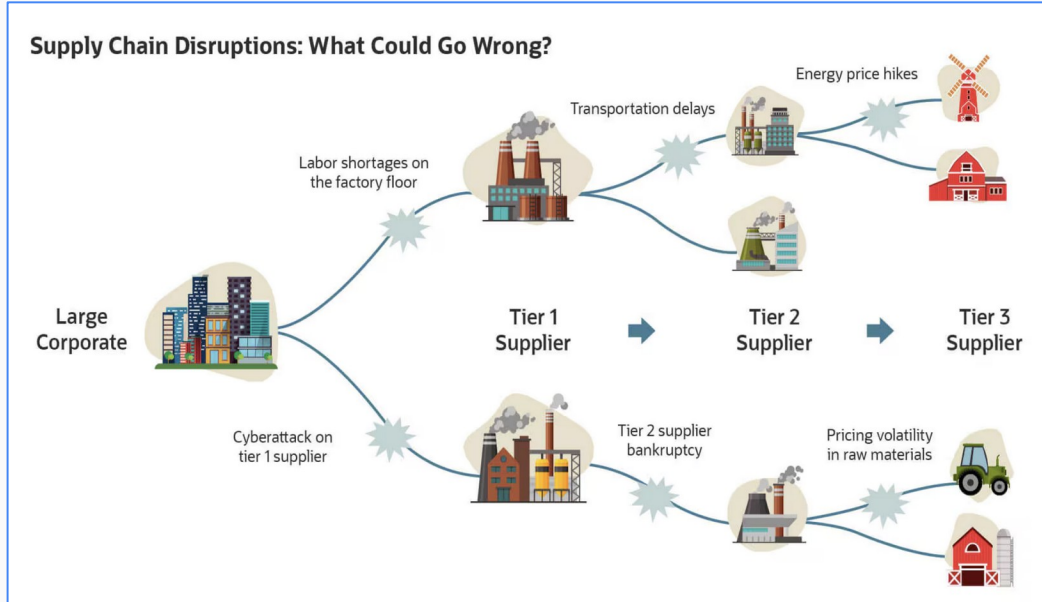
Five Key Elements

1. Risk Identification

Companies must identify potential risks in their supply chains. This includes understanding which suppliers, factories, or transportation routes are vulnerable to natural disasters.

2. Supplier Diversification

Companies should diversify their suppliers. Relying on only one supplier in a single region can be very risky. By working with multiple suppliers in different locations, businesses can continue production even if one supplier is affected by a disaster.



Five Key Elements Cont.

3. Inventory and Capacity Buffers

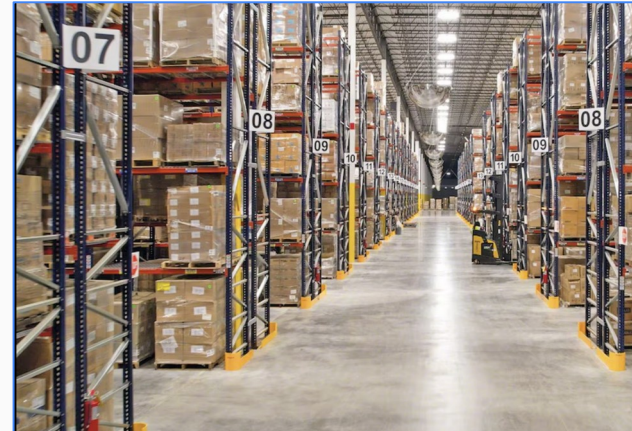
Companies should maintain safety stock and backup production capacity. Extra inventory and alternative production facilities can help companies continue operations when disruptions occur.

4. Digital Visibility

Technology plays an important role. Digital tools and real-time data systems allow companies to monitor their supply chains and respond quickly when problems arise.

5. Recovery and Communication plans

Companies need clear recovery plans and strong communication with suppliers and partners. A well-prepared recovery strategy helps businesses restore operations faster after a disaster.



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